PRACTICE EXAM 23 / EXAM SET 04

Domain 2: Anatomy and Physiology

8 scored questions	4 answer choices per question	Answer key + explanations included

Instructions

Choose the single best answer (A, B, C, or D) for each question. Mark your responses before checking the answer key and detailed explanations at the end of this document. These practice questions are not official NHA examination items.

QUESTIONS

Q01 | Body Structures and Organ Systems

> **SCENARIO** During a routine visit, a patient practices a provider-recommended breathing exercise and asks whether the diaphragm relaxes to pull air into the lungs. The medical assistant uses an approved diagram to explain normal quiet inhalation.

Which description correctly explains the diaphragm's role during inhalation?

A. It contracts and moves downward, increasing the volume of the chest cavity.
B. It relaxes and moves upward, increasing the volume of the chest cavity.
C. It contracts and moves upward, decreasing the volume of the chest cavity.
D. It relaxes and moves downward, decreasing the volume of the chest cavity.

Q02 | Pathophysiology and Disease Processes

> **SCENARIO** A patient with provider-confirmed obstructive sleep apnea is reviewing the sleep-study summary. The patient asks why the report uses the word "obstructive" when describing the nighttime pauses in airflow. The medical assistant reinforces the provider's explanation.

Which mechanism is characteristic of obstructive sleep apnea?

A. The brain intermittently stops sending the signals needed to initiate breathing.
B. The lower airways repeatedly narrow because of allergic inflammation in the lungs.
C. Upper-airway tissues narrow or block the passage of air during sleep.
D. Fluid in the pleural space prevents the lungs from expanding during sleep.

Q03 | Body Structures and Organ Systems

> **SCENARIO** A child returns after a throat infection. The provider documents reactive cervical lymph nodes and explains that these structures participate in the local immune response. The parent asks whether these nodes mainly filter the blood.

Which description correctly states the normal role of these lymph nodes?

A. They filter circulating blood and remove aging red blood cells.
B. They produce digestive enzymes that enter the upper throat.
C. They produce the hormones that regulate the body's metabolic rate.
D. They filter lymph and contain immune cells that respond to foreign material.

Q04 | Pathophysiology and Disease Processes

> **SCENARIO** A patient with provider-diagnosed acute viral hepatitis has yellowing of the skin and whites of the eyes. The provider has explained that impaired liver processing is allowing a pigment to accumulate. The medical assistant reinforces this explanation.

Which substance is responsible for this yellow discoloration?

A. Hemoglobin
B. Bilirubin
C. Melanin
D. Keratin

Q05 | Pathophysiology and Disease Processes

> **SCENARIO** At a scheduled follow-up, a patient with right-sided heart failure has stable bilateral ankle edema that the provider attributes to the heart condition. The patient asks how a problem in the chest can cause swelling in the legs. The medical assistant reinforces the documented explanation.

Which mechanism best connects the impaired right-heart pumping with this edema?

A. Reduced liver albumin production lowers the force retaining water within blood vessels.
B. Inflammation in the ankle joints increases local tissue swelling around the joint capsule.
C. Obstruction of leg lymphatic vessels prevents removal of interstitial fluid.
D. Blood backs up in systemic veins, promoting movement of fluid into surrounding tissues.

Q06 | Body Structures and Organ Systems

Which normal function is performed by platelets during the initial response to a small blood vessel injury?

A. Transporting oxygen from the lungs to injured tissues.
B. Engulfing bacteria that enter through the injured surface.
C. Adhering at the damaged site and aggregating to form an initial plug.
D. Maintaining plasma oncotic pressure to retain water in blood vessels.

Q07 | Body Structures and Organ Systems

SCENARIO A patient with provider-documented mitral valve regurgitation is reviewing a heart diagram during follow-up. The provider has described the valve as not closing tightly. The medical assistant reinforces the valve's normal function.

Which normal function is reduced when a heart valve does not seal properly?

A. Generating the electrical impulse that starts each heartbeat.

B. Maintaining one-way blood flow by preventing backward leakage.

C. Adding oxygen to blood before it enters the left ventricle.

D. Filtering metabolic wastes from blood before it enters the aorta.

Q08 | Pathophysiology and Disease Processes

SCENARIO A patient attends an outpatient follow-up after hospitalization for a provider-diagnosed ischemic stroke. The patient says, "I thought every stroke meant that a blood vessel burst." The medical assistant reinforces the discharge explanation about stroke types.

Which process describes an ischemic stroke?

A. An arterial blockage reduces blood flow and oxygen delivery to part of the brain.

B. A ruptured cerebral blood vessel releases blood into or around brain tissue.

C. Abnormal electrical activity spreads through brain cells and produces a seizure.

D. Degeneration of peripheral sensory nerves reduces sensation in the feet.

ANSWER KEY & EXPLANATIONS

Practice Exam 23 | Domain 2: Anatomy and Physiology | 8 scored questions

01 A	02 C	03 D	04 B	05 D	06 C
07 B	08 A				

DETAILED EXPLANATIONS

Q01 | Correct answer: A

It contracts and moves downward, increasing the volume of the chest cavity.

During quiet inhalation, the diaphragm contracts and moves downward, increasing thoracic volume. The resulting pressure change allows air to flow into the lungs. Relaxation and upward movement of the diaphragm normally contribute to quiet exhalation, linking skeletal muscle function with ventilation.

Q02 | Correct answer: C

Upper-airway tissues narrow or block the passage of air during sleep.

Obstructive sleep apnea involves repeated narrowing or blockage of the upper airway during sleep. Problems with the brain's respiratory signals characterize central sleep apnea, a different mechanism. Lower-airway allergic narrowing and pleural fluid do not describe the obstructive process documented here.

Q03 | Correct answer: D

They filter lymph and contain immune cells that respond to foreign material.

Lymph nodes filter lymph and contain immune cells, including lymphocytes, that help respond to infection and other foreign material. They are different from the spleen, which filters blood and removes aging red blood cells. The existing provider assessment explains the context; the assistant is not determining the cause of an enlarged node.

Q04 | Correct answer: B

Bilirubin

Bilirubin is a yellowish substance produced during the breakdown of old red blood cells and normally processed and eliminated by the liver. Accumulation can cause jaundice, including yellowing of the skin and sclerae. Melanin contributes to normal skin pigmentation but is not the pigment responsible for jaundice.

Q05 | Correct answer: D

Blood backs up in systemic veins, promoting movement of fluid into surrounding tissues.

Impaired right-heart pumping can cause blood to back up in the systemic venous circulation. Increased vascular pressure promotes movement of fluid into surrounding tissues, producing peripheral edema. Low albumin, local inflammation, and lymphatic obstruction can also cause swelling, but they do not describe the right-heart mechanism specified in this record.

Q06 | Correct answer: C

Adhering at the damaged site and aggregating to form an initial plug.

Platelets adhere to a damaged vessel and aggregate to help form the initial platelet plug. Clotting factors subsequently help stabilize the plug with fibrin. Red blood cells carry oxygen, certain white blood cells engulf pathogens, and plasma proteins such as albumin help maintain oncotic pressure.

Q07 | Correct answer: B

Maintaining one-way blood flow by preventing backward leakage.

Heart valves help maintain one-way blood flow by opening to allow forward movement and closing to prevent backflow. A valve that does not close tightly can allow regurgitation, or backward leakage. Generating the heartbeat's initiating impulse is a function of the cardiac conduction system rather than the valve.

Q08 | Correct answer: A

An arterial blockage reduces blood flow and oxygen delivery to part of the brain.

An ischemic stroke results from blocked blood flow that deprives part of the brain of oxygen and nutrients. A hemorrhagic stroke results from bleeding into or around the brain after a vessel ruptures. The question concerns an established diagnosis and does not ask the assistant to distinguish stroke types in a patient with new symptoms.

End of Practice Exam